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Paleomagnetic results from the Jurassic La Quinta Formation in the Perijá Range, Venezuela, and their tectonic significance

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ABSTRACT

The paleomagnetism of 150 oriented hand samples (32 sites) of volcanic, red bed, and dike rocks, constituting the La Quinta Formation in the Perijá Range of Venezuela, has been studied. After demagnetization, consistent vectors from volcanic rocks and red beds are bipolar but show considerable scatter both before and after correction for present dip. A northwest-southeast declination predominates, however, in contrast to the northeast-southwest declination previously recorded in La Quinta strata from the Venezuelan Andes. As the declination predicted from data from the South American craton is northerly, differential rotation betwen the Perijá Range and Mérida Andes, causing opening to the north, is suggested. Six dikes give consistent northerly vectors, indicating emplacement after the differential rotations, probably in Early Cretaceous time.

INTRODUCTION

The La Quinta Formation in the Perijá Range of northwestern Venezuela consists of a sequence of continental sedimentary and igneous rocks 1,000 to 1,700 m thick (Maze, 1984). Red arkosic sandstone and siltstones are the dominant sedimentary rocks, with some conglomerate as well as minor occurrences of limestone and shale. The igneous rocks, which in the field area constitute as much as 30% of the La Quinta exposures, may be loosely divided into three categories: (1) flow rocks, (2) volcaniclastic rocks, and (3) dikes. Two types of flow rocks are distinguished: vesicular basaltic-andesite flows 1 to 3 m thick and nonvesicular porphyritic hornblende-andesite flows 5 to 15 m thick. Flow-banding is a common feature of the hornblende-andesite. Volcaniclastic rocks interbedded with the sedimentary and flow rocks are mostly rhyolitic and rhyodacitic and occur as tuffs, welded tuffs, and debris flows from centimeters to hundreds of meters thick. Flattened clasts and flow-banding are indicators of bedding which, however, may not truly reflect a paleohorizontal attitude. Dikes cut all rock types within the La Quinta Formation. All dikes sampled for this study are alkalic basalts with chemistry distinctly different from that of the flow and volcaniclastic rocks.

The paleontologic age of the La Quinta Formation has been established as Middle to Late Jurassic by Benedetto and Odreman (1977). U-Pb, Rb-Sr, and K-Ar isotopic ages on the volcanic rocks range from about 170 to 140 m.y. (Espejo and others, 1980; Dasch, 1982; Steinitz and Maze, 1984; Maze, 1984). Red bed formations of generally similar lithology and age are known also in the Venezuelan Andes (the type locality for the La Quinta Formation; Hargraves and Shagam, 1969), on the Colombian side of the Perijá Range and at Santander (Giron Formation; Cediel, 1969), the Guajira Peninsula (MacDonald and Opdyke, 1972), and Isla de Toas (Pimentel, 1975; Fig. 1).

Oriented samples were collected in the hope that paleomagnetic data might (a) aid in local stratigraphic correlations within the Perijá Range and (b) elucidate the structural history of the Perijá-Maracaibo-Andes block by comparison with results from the La Quinta in the Venezuelan Andes (Hargraves and Shagam, 1969) and rocks of similar age elsewhere in cratonic South Amer-

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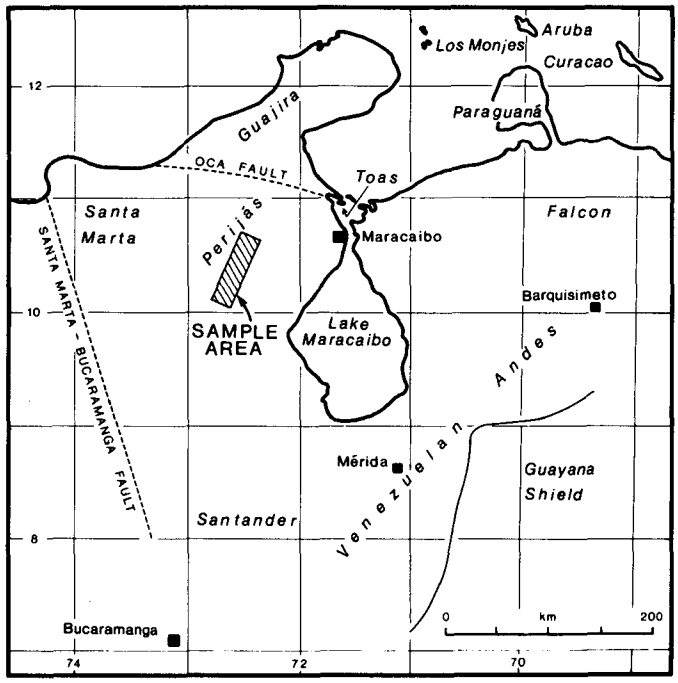


Figure 1. Map showing area sampled in Perijá Range and location of Toas, northwestern South America.

ica. The results, presented below, were of no value with respect to local correlation, but do provide evidence of differential tectonic rotation between the Perijá Range and Andes.

SAMPLING AND MEASUREMENT

One hundred and fifty oriented hand samples were collected from 32 sites as shown in Figures 1 and 2. The samples include red sedimentary rocks as well as mafic to intermediate or felsic intrusive and extrusive rocks (see Table 1). Standard 2.5-cm cores were drilled from these samples in the laboratory.

Paleomagnetic study included measurement of natural remanent magnetism (NRM) (Schonsted DSM 1) and stepwise alternating field (AF) demagnetization of selected specimens to 70 milliteslas (mT) (700 Oe) using a 2-axis tumbler. Thirty-two specimens were thermally demagnetized in a field <20 nanoteslas (nT) (≦20 gamma) to 650 °C. These data were analyzed by vector subtraction and plotting of Zijderveld (1967) diagrams (Fig. 3) whereby single-component vectors could be identified. Site-mean results reported in Table 1, however, are those showing minimum dispersion at a specified optimum demagnetizing field (ODF) or temperature (ODT).

PETROLOGY AND MAGNETIC PROPERTIES

Volcanic Rocks

The basaltic andesites are dark, fine-grained rocks occurring as flows with vesicular tops and brecciated bottoms; the

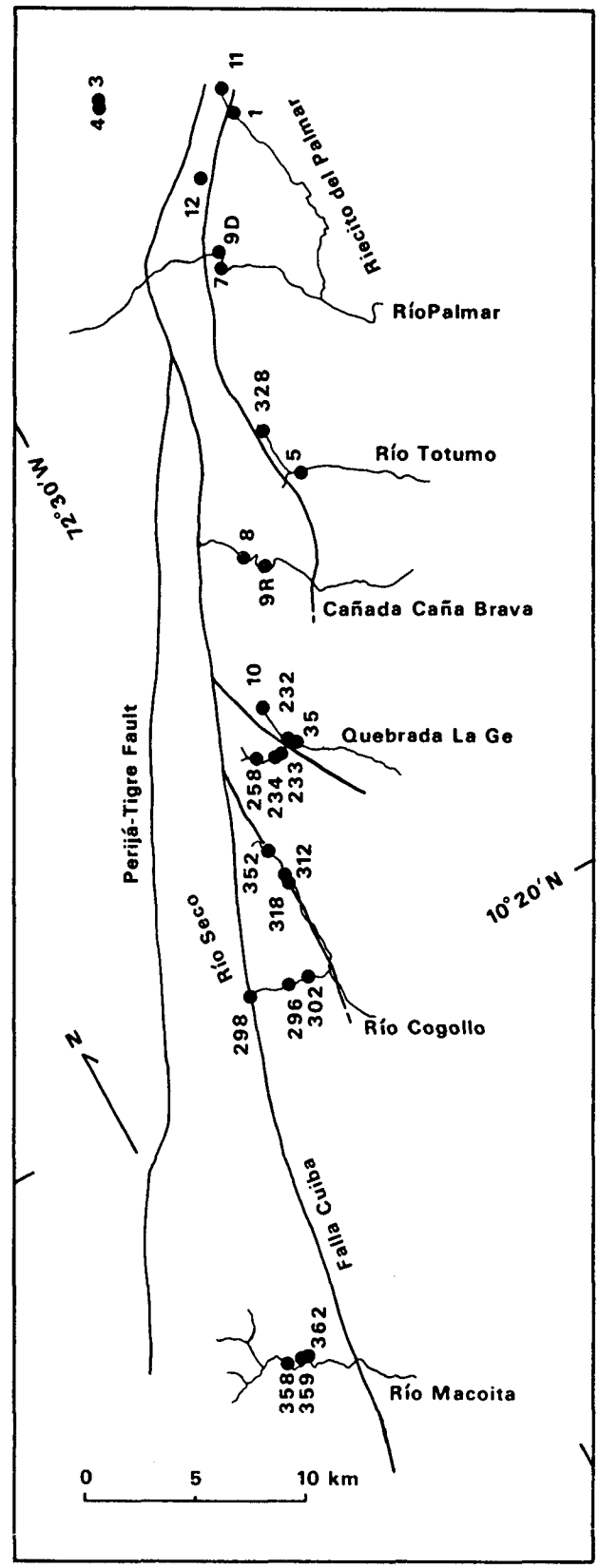


Figure 2. Map showing sampling site locations. Locations for sites 3, 4, 11, and 12 are approximate.

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TABLE 1. PALEOMAGNETIC MEASUREMENTS ON PERIJÁ RANGE SAMPLES UNCORRECTED FOR BEDDING

			NRM				Demagnetized					
Site No.	Petrology	N	D_n	I_n	k	$\bar{J}_n$	MDF_{mT}	ODF/T	D	I	k	α_{95}
	Volcanic rocks											
3	agglomerate	4	290	-9	13	$1x10^{-5}$	40	20	290	-1	9	32
4	volcaniclastic	4	213	-44	2	$3x10^{-6}$	40	20	179	-41	15	24
5	hornblende andesite	7(4)	347	25	20	$5x10^{-4}$	20	50	340	-2	148	8
7	dacite porphyry	5(4)	228	67	3	$1x10^{-6}$	50	50	137	57	44	12
8	basaltic andesite	5	328	39	2	$5x10^{-4}$	7.5	40	303	-9	14	21
11	pyroclastic	3	257	69	13	$1x10^{-5}$	5	30	224	50	12	37
12	vesicular lava	5	325	15	6	$1x10^{-4}$	5	30	334	5	8	29
35	rhyolite tuff	5	123	57	12	$4x10^{-7}$	30	30	150	38	32	14
232	rhyolite tuff	5	142	-25	105	$1x10^{-4}$	40	40	143	-26	114	7
233	hornblende andesite	9	87	39	2	$1x10^{-4}$	40	30	130	-26	9	19
258	basaltic andesite	5	154	25	114	$5x10^{-5}$	50	30	155	23	121	7
298	rhyolite tuff	5	304	70	9	$1x10^{-6}$	20	20	293	58	12	23
328	volcaniclastic	4	143	28	140	$8x10^{-5}$	40	20	146	20	169	7
	Dikes											
1	alkali basalt	5	341	40	2	$1x10^{-4}$	7.5	20	357	43	11	24
7	alkali basalt	7	349	47	5	$4x10^{-4}$	20	30	350	27	65	8
9D	alkali basalt	4	12	61	11	$5x10^{-4}$	20	20	357	26	16	23
302	alkali basalt	4	348	37	8	$6x10^{-4}$	7.5	30	343	25	10	31
312	alkali basalt	2	354	45	651	$2x10^{-4}$	6	20	00	49	229	17
352	alkali basalt	4	06	44	23	$3x10^{-4}$	12.5	30	04	42	12	28
Toas	basalt	4	328	46	6	$9x10^{-4}$	50	30	350	37	56	17
	Red Beds						MDT°C					
4		2	06	19	2	$8x10^{-6}$	300°	400°	343	-1	69	31
9R		4(3)	339	42	10	$6x10^{-6}$	250°	500°	300	-5	5	62
10		5	330	60	10	$2x10^{-5}$	200°	555°	261	13	10	26
234		6	00	57	4	$1x10^{-5}$	100°	305°	27	45	7	29
296		4	01	43	16	$9x10^{-5}$	150°	600°	75	39	17	23
302		4	347	23	24	$2x10^{-5}$	450°	600°	334	-9	51	13
312		2	346	59	89	$2x10^{-5}$	75°	200°	328	41	4	90
318		3	328	25	9	$6x10^{-5}$	300°	530°	303	6	39	20
352		5	15	53	44	$3x10^{-6}$	400°	555°	37	58	106	8
358		5	348	54	4	$1x10^{-5}$	150°	500°	24	14	5	36
359		5	359	36	16	$1x10^{-5}$	250°	400°	29	23	8	29
362		5	344	51	43	$7x10^{-6}$	150°	500°	348	67	2	85

Note: Site number and location are indicated on Figure 2; N = number of samples collected and used in analysis. Smaller numbers in parentheses, where indicated, were used in ODF/ODT analyses. D, I = magnetic decliniation and inclination (+ down); subscript n refers to original NRM; k and α_{95} are Fisher's statistical parameters. $\bar{J}_n$ is the average intensity (emu/cm^3) or NRM. MDF(mT) or MDT(°C) are the mean demagnetizing field or temperature, which is the average alternating field (mT) or temperature (°C) required to reduce NRM to half of its original intensity. ODF/T is the optimal demagnetizing field or temperature, giving the best site-mean statistics. Whole-rock K-Ar ages on hornblende andesite from sites 5 and 233 are 155 ± 5 and 146 ± 7 m.y., respectively (Steinitz and Maze, 1984). U-Pb ages on zircons from the hornblende andesite at site 5 are 163 ± 5 m.y. (Dasch, 1982). These zircons gave a 90 ± 9 m.y. fission-track age (Shagam and others, 1984).

hornblende-andesite flows are coarser grained (porphyritic) and lack vesicular tops. Clinopyroxene, plagioclase, Fe-Ti oxides, and apatite are present in both, but amphibole occurs only in the latter. Although the degree of alteration varies considerably from site to site, the feldspars in the more mafic rocks are somewhat saussuritized, whereas the pyroxenes and amphiboles are usually fresh. Conspicuous hematite-rich coronas occur around hornblende phenocrysts. Primary opaques are relict titanomagnetites, with both ilmenite lamellae and host spinel variably replaced by hematite. Little evidence of maghemite was seen. The mineralogy and texture suggest that the alteration is deuteric, occurring at high temperatures during the cooling of these mafic lavas rather than much later at lower temperatures.

The felsic volcanic rocks commonly are pervasively altered and silicified, with a few wispy chloritic relicts remaining after original ferromagnesians. Original clinopyroxene and amphibole can be identified in the least-altered rocks, but there is little evidence of primary oxide phases.

With the exception of some of the more fragmental volcaniclastics and those felsic lavas more intensely altered by hy-

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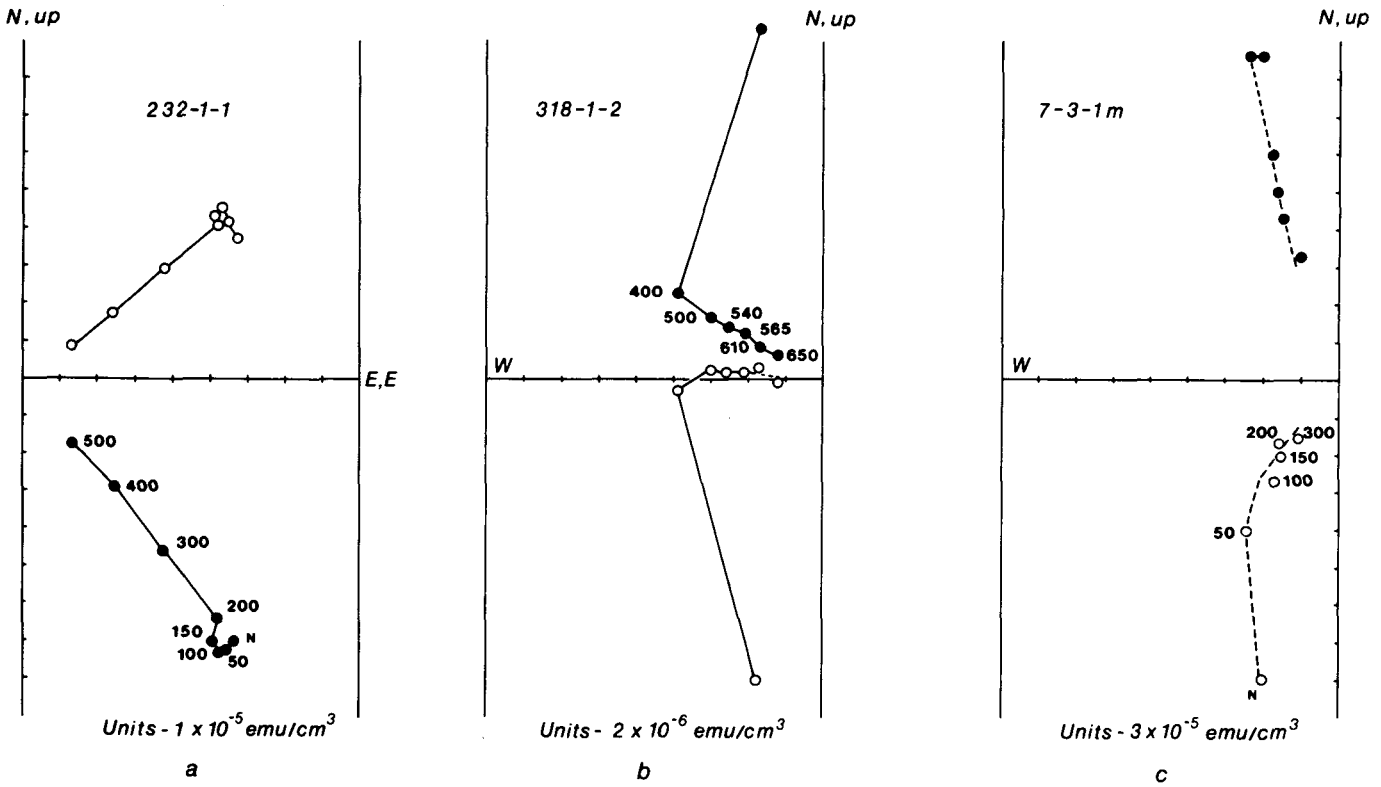


Figure 3. Zijderveld plots illustrating demagnetization of various rock types, in situ coordinates; rhyolite tuff (a); red bed (b); dike rock (c).

drothermal activity, the median demagnetizing field (MDF) of all other volcanic rocks is high (20 to 50 mT, see Table 1), suggesting that single-domain magnetite or hematite is the carrier of the remanent magnetism. This is consistent with the observation of hematite rimming and veining the primary spinels and as an alteration product of hornblende in some samples. Radiometric ages of the hornblende andesite at site 5 and site 233 are given in Table 1.

Sedimentary Rocks

The red beds sampled are medium- to fine-grained, well-bedded red or red-brown to purple arkosic sandstones and siltstones. AF demagnetization of some samples (during which little change in NRM direction occurred) revealed a MDF of between 10 and 20 mT. On thermal demagnetization, most samples showed a steady decline to between 0.5 and 0.2 of the initial moment (J_N) at 500 or 525 °C, followed at higher temperatures by more irregular behavior. Most samples showed a continuing steeper drop in intensity after 500 or 525 °C, but some showed a slight recovery between 560 and 600 °C. The steeper drop in intensity above 500 °C to 530 °C is not accompanied by any significant change in direction, which remained constant in individual samples even after demagnetization to 650 °C; the within-site grouping, however, was only fair (see Table 1). The site mean result showing minimum dispersion at one of the 500, 530, 560, 600, or 650 °C demagnetization steps is reported in Table 1.

Dikes

The basaltic dikes, varying from 0.3 to 5 m in thickness, are in general more equigranular and appear finer grained than the volcanic rocks. Alteration has affected the plagioclase and oxide grains and caused minor chloritization of the pyroxenes; however, the pyroxene microphenocrysts are usually quite fresh. As in the volcanic rocks, the larger opaque grains in the dikes show evidence of high-temperature oxyexsolution of ilmenite from magnetite followed by intense alteration of the ilmenite lamellae and partial replacement of the relict spinel by hematite.

The remanent magnetism in all seven dikes sampled is constant in direction and of intermediate stability with respect to AF demagnetization (MDF 6 to 20 mT).

SUMMARY AND DISCUSSION OF PALEOMAGNETIC RESULTS

The site means of the best single-component vectors that have been isolated by systematic demagnetization of all samples are listed in Table 1 and illustrated in Figures 4 and 5.

It is apparent that with or without correction for bedding

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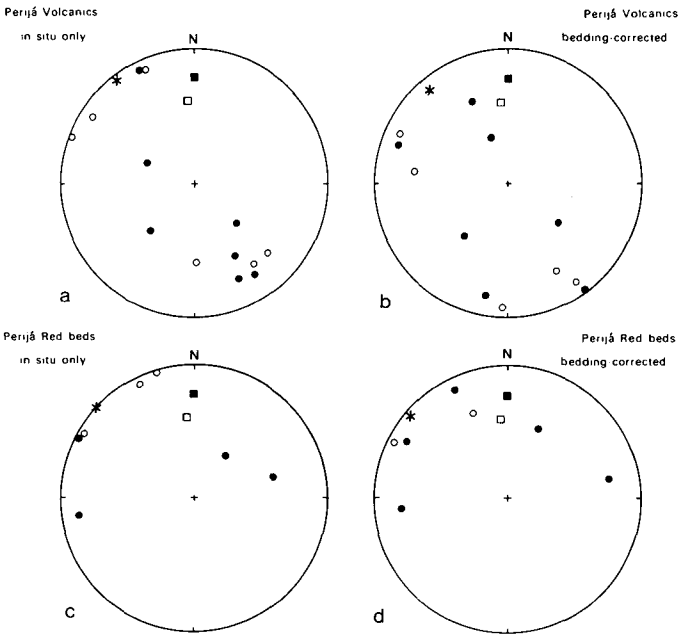


Figure 4. Equal-area projections of demagnetized vectors from volcanic rocks (a, b) and red beds (c, d) before and after structural (bed) correction (when structural correction is possible). Open circles are upper hemisphere, and solid circles are lower hemisphere. Open square represents present Earth's field, and solid square is present theoretical dipole field. Statistics are in Tables 1 and 2. Means from Table 2 plotted as *.

there is considerable scatter in vector directions for both red beds and volcanic rocks. The dikes, on the other hand, are well grouped.

Red Beds and Volcanic Rocks

Both normal and reverse magnetizations are evident in the volcanic rocks and possibly in the red beds (see Fig. 4). Positive and negative inclinations are associated with both polarities, but the average is nearly horizontal both before and after bed correction (Table 1). As the vectors are northwest-southeast, and the prevailing strike is northeasterly, structural corrections primarily affect the inclinations, but in general, where correction shallows one site vector, it steepens another, and the average remains approximately the same.

The volcanic-rock data (Fig. 4a, 4b), which include the magnetizations most likely to have been acquired during initial cooling, are the most consistent in orientation at individual sites. Unfortunately, the foliation interpreted as bedding in these units may not be dependable as an indicator of a paleohorizontal attitude, as discussed in the introduction. The scatter between site-mean vectors, therefore, may in part reflect errors in structural correction, as well as the errors to be expected from secular variation. Some of the volcanic units are intensely altered, and the magnetization in these may be entirely secondary. As tectonism associated with volcanism probably occurred during the deposition of the La Quinta Formation (and tectonism has certainly occurred since), the secondary magnetizations, even if occurring soon after deposition or eruption, may be of arbitrary orientation.

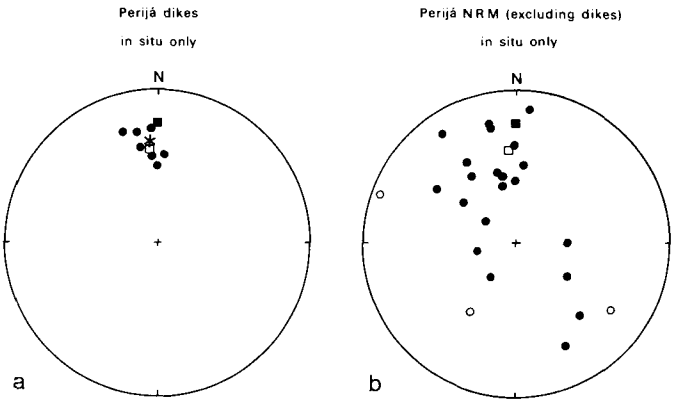


Figure 5. Equal-area projections of demagnetized dike vectors (a) and NRM of all Perijá sites (uncorrected for bedding attitude) (b). Dike mean from Table 2 is plotted as *. Symbols as in Figure 4.

Bedding in the red beds is quite clear, but the timing of magnetization in such strata is notoriously uncertain (Larson and others, 1982). Given the turbulent geologic history of this area since Jurassic time, the scatter in these data is not surprising, even if it can be assumed that single-component magnetizations have been isolated.

Dikes

The six dikes sampled are comparatively well grouped (Fig. 5a and Table 1), in situ D = 354, I = 36, k = 47. Country-rock bedding attitude is known only for three of the dikes (sites 302, 312, and 352, Fig. 2), and correction for this flattens the inclination somewhat (N = 3, D = 06, I = 27, k = 33), but it remains relatively high. A dike cutting granite on Isla de Toas (Fig. 1) has a similar NRM (Table 1). This mean vector is close to that of the theoretical dipole field for Maracaibo at the present time (D = 360°, I = +20°). The stability of the dike vectors to alternating fields in excess of 20 mT, however, convinces us that the magnetization is primary and not a modern-day overprint.

The absolute age of the dikes is not known. Petrochemically, they are distinctly alkaline (Maze, 1984) and hence unlike any of the La Quinta volcanic rocks which they intrude. They could thus be considerably younger than the La Quinta strata. The absence of any evidence of igneous activity in the Maracaibo area since the beginning of Cretaceous sedimentation, however, suggests that they are no younger than Early Cretaceous, and possibly they are directly related to the closing stages of La Quinta volcanism.

Thermal Overprint

The in situ site mean NRM directions (predemagnetization) of all La Quinta samples measured in this study are projected in

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TABLE 2. PALEO-DECLINATION, INCLINATION, AND LATITUDE OF MARACAIBO RELATIVE TO PRESENT THEORETICAL DIPOLE FIELD (a); MEAN DECLINATIONS AND INCLINATIONS OF PERIJÁ ROCKS (b)

(a) Time m.y.	Declination°	Inclination°	Latitude
180	13°E	+4	2°N
160	13°E	+10	5°N
140	13°E	+6	3°N
120	7°E	+6	3°N
100	0	+10	5°N
80	10°W	+14	7°N
60	10°W	+14	7°N
40	1°W	+10	5°N
20	3°W	+8	4°N
Present theoretical dipole field	0°	+20	10°30'N
Present field direction	4.5°W	+39	--

(b) Mean Directions after ODF or ODT	D°	I°	Confidence angles	M.L.E.(K1/K2)
			Bingham Statistics	
Volcanic Rocks				
in situ (12)	323	-2	13.5/28.4	-7.4/-2.3
bedding corrected (10)	319	+9	16.4/35.9	-6.5/-2
Red Beds				
in situ (5)	311	0	6.3/36.1	-65.3/-2.7
bedding corrected (5)	309	+3	15.1/35.7	-12.3/-2.9
Perijá Dikes				
in situ (6)	354	36	3.3/10.4	-155.4/-16.2
bedding corrected (3)	6	27	4.2/16	-192.2/-13.9

Note: (a) As inferred from paleogeographic history of cratonic South America (Barron and others, 1981). Present field direction from Magnetic Variation Epoch 1975.0 and Magnetic Inclination or Dip Epoch 1975.0 maps prepared and published by the Defense Mapping Agency Hydrographic Center, Washington, D.C. 20390. (b) From measurements of La Quinta rocks in the Perijá Range after ODF or ODT demagnetization, calculated using Bingham statistics (Onstott, 1980). Number of samples used in calculating the mean given in parentheses. Samples with low Fisher statistics and/or unreliable geologic field relationships were excluded. Bedding control known for only 3 Perijá dikes. The two confidence angles are a measure of the ellipticity of the vector probability distribution (cf. α_{95}) and the MLE is the "maximum likelihood estimate", analogous to Fisher's k (see Onstott, 1980).

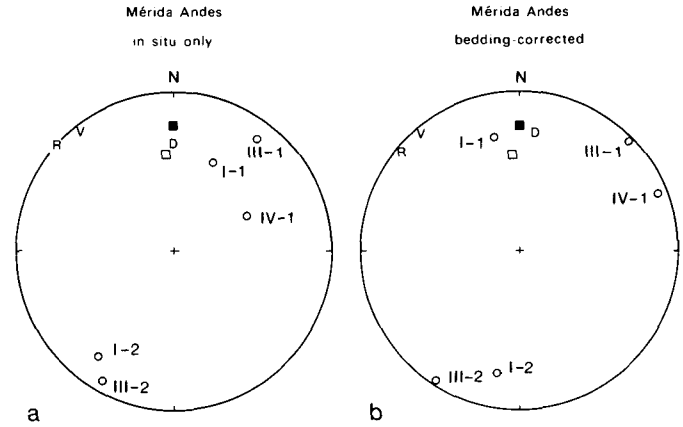


Figure 6. Equal-area projections of La Quinta red-bed normal and reverse group means from three of the four sites studied in detail in the Mérida Andes before correction (a) and after correction (b) for bedding attitude (Hargraves and Shagam, 1969, Fig. 10). Omitted are vectors shown in the original figure from the Permian-Triassic tuffs (III-1) and from locality V, now considered to have undergone intense brecciation (see Giegengack, 1984). Perijá means for red beds, volcanic rocks, and dikes from Table 2 are plotted as R, V, and D, respectively. Symbols as in Figure 4.

Figure 5b. A similarity to the mean-dike vector is apparent, and indeed, the vectors erased during early stages of both AF and thermal demagnetization trend northerly, in general, with moderately steep positive inclination (e.g., Fig. 3b, 3c). This suggests the pervasive regional presence of a viscous overprint parallel to that of the dikes and presumably acquired at the time of their emplacement.

TECTONIC SIGNIFICANCE

The disruption of Pangea, starting in Jurassic time, involved the separation of South America from both Africa and North America. Spreading associated with the formation of the South Atlantic and the Caribbean required subduction on the Pacific margins. The northwestern border of the Guayana shield (Fig. 1) must have borne the brunt of the stresses and deformations associated with these events. The Perijá Range is located in the extreme northwestern segment of South America, much of which was deformed, modified, or possibly accreted during this interval.

As a baseline to which our results should be compared, we have determined the magnetic declination and inclination of Maracaibo, as a function of time, from the best published estimates of the paleogeography of cratonic South America (Table 2; Barron and others, 1981). Although the near-horizontal average inclination of the Perijá La Quinta strata (Table 2) is consistent with that predicted, the northwesterly declination is distinctly anomalous. Counterclockwise rotation of at least a large part of the Perijá Range is indicated.

The northerly declination of the dikes is consistent with that predicted from the craton, which suggests that the counterclockwise rotation of the Perijá Range was completed by the time of dike emplacement. The mean inclination of the dikes, however, is anomalously steep for times older than 20 m.y. (Table 2), implying a paleolatitude of at least 13°N at the time of intrusion. Alternatively, large blocks of the Perijá Range could have been tilted northward since Cretaceous time, but for this there is no independent evidence. Gose and Swartz (1977) have determined a transient excursion of Honduras 20° north of its present latitude in the Albian, which might be related to the dike inclination anomaly reported here.

The La Quinta paleomagnetic record in the Mérida Andes, reported previously (Hargraves and Shagam, 1969), was likewise highly variable and deemed in large part to be secondary. The only unique feature there was a prevailing northeasterly declination (Fig. 6), for which tectonic rotation was invoked. The rotations implied by the La Quinta paleomagnetic data in the Mérida Andes (~0–60° clockwise) and the Perijá Range (20°–60° counterclockwise) are thus in opposite directions. The implied opening in Late Jurassic or earliest Cretaceous time may have set the stage for the Maracaibo basin structurally, but the thick accumulation of sediment began only in Eocene time. This resulted from erosion of a source area developed to the northeast after collision with an island arc in latest Cretaceous time (Beets and others, 1984). The predominant present and recent tectonic style in the ranges surrounding the Maracaibo basin, however, is overthrusting to the northwest: a Tertiary compressive regime incompatible with the Jurassic tension required by this study. The La Quinta

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strata themselves were deposited in a series of graben, and perhaps the rotations indicated by the paleomagnetic data are part of this transient episode, heralding the thick accumulation of sedimentary strata in the Maracaibo basin.

The paleomagnetic data reported here, however, are few and of uneven quality; they can at best be considered to suggest the exceedingly complex tectonic history experienced since Jurassic time by all terranes on the northwestern margin of South America.

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